

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total(10)
<i>Weightage</i>	20%	25%	25%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I Ch.Sathish/(192311032)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Ch.Sathish

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation
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Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1

User Product Views Clicks Purchases

U5	55	22	5
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Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

ANSWERS

Q1. Hospital Patient Segmentation – K-Means and Elbow Method

Step 1: Understand the objective

*The healthcare organization wants to group patients according to their **monthly medical expense and health risk score.***

K-Means clustering divides patients into groups where patients within the same group are more similar to each other.

Step 2: Observe the SSE values

<i>Number of Clusters (K)</i>	<i>SSE</i>
<i>1</i>	<i>600</i>
<i>2</i>	<i>350</i>
<i>3</i>	<i>220</i>
<i>4</i>	<i>200</i>
<i>5</i>	<i>190</i>

SSE (Sum of Squared Errors) measures how closely the data points are grouped around their cluster centers. A lower SSE indicates better compactness.

Step 3: Calculate the reduction in SSE

- $K = 1 \rightarrow 2: 600 - 350 = 250$*
- $K = 2 \rightarrow 3: 350 - 220 = 130$*

- $K = 3 \rightarrow 4: 220 - 200 = 20$
- $K = 4 \rightarrow 5: 200 - 190 = 10$

Step 4: Identify the elbow

The SSE decreases significantly until $K = 3$. After $K = 3$, the improvement becomes very small.

Therefore, the **elbow point is $K = 3$** . Step

5: Select the optimal clusters Optimal number of clusters = 3.

The patients can be segmented into three broad groups:

- **Cluster 1:** Lower medical expenses / lower-to-moderate risk
- **Cluster 2:** Moderate medical expenses / higher risk
- **Cluster 3:** Higher medical expenses / different risk requirements The exact cluster membership would depend on the actual K-Means calculation.

Step 6: Effect on healthcare decision-making Three clusters allow hospitals to:

1. Identify high-risk patients.
2. Provide preventive healthcare to vulnerable groups.
3. Allocate doctors and medical resources efficiently.
4. Develop different treatment strategies.
5. Identify high-cost patient groups.
6. Improve healthcare planning and budgeting.

Final Conclusion

The elbow method indicates that $K = 3$ is the optimal number of clusters because the SSE reduction becomes very small after three clusters. This gives meaningful patient segmentation without creating unnecessary clusters.

Q2. E-Commerce Order Analysis – Hierarchical Clustering

Step 1: Understand the objective

The retailer wants to group orders according to:

- Product category
- Order value

Hierarchical clustering creates a tree-like structure called a **dendrogram**.

Step 2: Examine the orders

Order	Category	Value
O1	Electronics	₹55,000
O2	Clothing	₹25,000
O3	Electronics	₹57,000

<i>O4</i>	<i>Furniture</i>	<i>₹30,000</i>
<i>O5</i>	<i>Clothing</i>	<i>₹27,000</i>

Step 3: Identify similar orders *O1* and *O3* are highly similar because:

- Both are Electronics.
- Their values are ₹55,000 and ₹57,000.

Therefore:

O1 + O3 → Cluster 1

Similarly, *O2* and *O5* are similar:

- Both are Clothing.
- Their values are ₹25,000 and ₹27,000.

Therefore:

O2 + O5 → Cluster 2

O4 is Furniture and has a different category.

O4 → Cluster 3

Step 4: Interpret the dendrogram In hierarchical clustering:

1. Initially, each order is an individual cluster.
2. The most similar orders are merged first.
3. Further similar clusters are merged.
4. A horizontal cut in the dendrogram determines the final number of clusters.

A suitable interpretation for this dataset is **3 clusters**.

Step 5: Final clusters

Cluster	Orders	Interpretation
<i>C1</i>	<i>O1, O3</i>	<i>High-value Electronics</i>
<i>C2</i>	<i>O2, O5</i>	<i>Clothing / medium-value orders</i>
<i>C3</i>	<i>O4</i>	<i>Furniture</i>

Step 6: Business significance The clusters help the retailer:

1. Identify product preferences.
2. Understand spending patterns.
3. Create category-specific advertisements.
4. Recommend similar products.
5. Provide targeted discounts.
6. Improve inventory planning.

Final Conclusion

The hierarchical clustering suggests **three meaningful customer/order segments**. These clusters help the retailer understand purchasing behavior and develop targeted marketing strategies.

Q3. Mobile App User Behavior – Soft Clustering

Step 1: Understand the objective

The mobile application company wants to segment users according to:

- *Daily usage time*
- *Login frequency*
- *In-app purchase amount*

Step 2: Examine the data

<i>User</i>	<i>Usage</i>	<i>Logins</i>	<i>Purchase</i>
<i>U1</i>	<i>120</i>	<i>18</i>	<i>₹2,000</i>
<i>U2</i>	<i>60</i>	<i>5</i>	<i>₹5,000</i>
<i>U3</i>	<i>140</i>	<i>22</i>	<i>₹1,500</i>
<i>U4</i>	<i>50</i>	<i>4</i>	<i>₹6,000</i>
<i>U5</i>	<i>130</i>	<i>20</i>	<i>₹1,800</i>

Step 3: Identify behavioral patterns U1, U3 and U5:

- *High usage time*
- *High login frequency*
- *Relatively lower purchase amounts*

These users can be considered **highly active users**.

U2 and U4:

- *Low usage time*
- *Low login frequency*
- *High purchase amounts*

These users can be considered **low-activity but high-spending users**.

Step 4: Understand soft clustering

In soft clustering, a user is not necessarily assigned completely to only one cluster.

For example:

<i>User</i>	<i>Active User</i>	<i>High-Spending User</i>
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<i>U1</i>	<i>0.90</i>	<i>0.10</i>
<i>U2</i>	<i>0.20</i>	<i>0.80</i>
<i>U3</i>	<i>0.95</i>	<i>0.05</i>
<i>U4</i>	<i>0.10</i>	<i>0.90</i>
<i>U5</i>	<i>0.92</i>	<i>0.08</i>

Note: These probability values are illustrative because actual membership probabilities are not provided in the question. **Step 5: Interpret the probabilities** For example:

- *U1 has 90% membership in the active-user cluster.*
- *U2 has 80% membership in the high-spending cluster.*
- *U4 has 90% membership in the high-spending cluster.*

This gives more information than simply saying "U1 belongs to Cluster 1."

Step 6: Advantages of soft clustering

- 1. Users can partially belong to multiple groups.*
 - 2. It handles uncertain customer behavior.*
 - 3. It identifies users near cluster boundaries.*
 - 4. It supports personalized marketing.*
 - 5. It improves recommendation systems.*
 - 6. It helps predict changes in user behavior.*
- Step 7: Soft vs Hard clustering**

Hard Clustering	Soft Clustering
<i>User belongs to one cluster</i>	<i>User can have membership in multiple clusters</i>
<i>Simple interpretation</i>	<i>More flexible</i>
Hard Clustering	Soft Clustering
<i>Cannot represent uncertainty well</i>	<i>Represents uncertainty</i>

Less suitable for overlapping behavior

Suitable for complex customer behavior

Soft clustering is useful for mobile applications because user behavior can overlap between different groups. It provides a more realistic representation of customer behavior than hard clustering.

Q4. Agricultural Crop Monitoring – Normalization Step 1: Identify the variables

<i>Farm</i>	<i>Rainfall</i>	<i>Soil Fertility</i>	<i>Crop Yield</i>
<i>F1</i>	<i>300</i>	<i>8</i>	<i>85</i>
<i>F2</i>	<i>500</i>	<i>4</i>	<i>70</i>
<i>F3</i>	<i>400</i>	<i>6</i>	<i>80</i>
<i>F4</i>	<i>280</i>	<i>9</i>	<i>90</i>
<i>F5</i>	<i>520</i>	<i>3</i>	<i>65</i>

Step 2: Observe the scale difference The variables have different ranges:

- *Rainfall = 280–520*
- *Soil fertility = 3–9*
- *Crop yield = 65–90*

Rainfall has a much larger numerical scale.

Step 3: Clustering before normalization

If raw values are directly used in a distance-based algorithm, rainfall can dominate the distance calculation.

For example, a rainfall difference of 200 units can have much more influence than a soil fertility difference of 2 units.

Therefore, clusters may be formed mainly based on rainfall.

Step 4: Apply normalization

Normalization transforms variables to a common scale.

*One common method is **Min-Max normalization**:*

*This converts the values approximately into the range **0 to 1**.*

Step 5: Clustering after normalization After normalization:

- *Rainfall has appropriate influence.*
- *Soil fertility has appropriate influence.*

- Crop yield has appropriate influence.

Therefore, clusters represent the **combined agricultural characteristics** rather than being dominated by rainfall. **Step 6: Possible groups Group 1:**

- F1 and F4
- Higher soil fertility
- Higher crop yield
- Lower/moderate rainfall **Group 2:**

- F2 and F5
- Lower soil fertility
- Lower crop yield
- Higher rainfall **Group 3:**

- F3
- Intermediate characteristics

Step 7: Advantages of normalization 1.

Prevents one variable from dominating.

2. Makes variables comparable.
3. Improves distance calculations.
4. Produces more meaningful clusters.
5. Improves reliability of agricultural analysis.
6. Helps researchers make better farming decisions.

Final Conclusion

Normalization is important because it ensures that **rainfall, soil fertility and crop yield contribute fairly** to clustering. Therefore, the normalized clustering results are generally more reliable.

Q5. Employee Performance Evaluation – Silhouette Score Step 1: Understand the objective

The company wants to determine the best number of employee clusters based on:

- Productivity
- Attendance
- Skill assessment

Step 2: Given silhouette scores

Model	Silhouette Score
$K = 2$	0.48
$K = 3$	0.68
$K = 4$	0.52

Step 3: Understand the silhouette score The silhouette score measures:

- How close an employee is to employees in the same cluster.
- How far the employee is from employees in other clusters.

A higher silhouette score indicates better-defined clusters.

Step 4: Compare the models

$K = 2 \rightarrow 0.48$

This gives moderate cluster quality, but the separation between clusters is not the best.

$K = 3 \rightarrow 0.68$

This is the highest score. It indicates that the clusters are relatively compact and well separated.

$K = 4 \rightarrow 0.52$

*The score decreases compared with $K = 3$, meaning that adding another cluster does not improve the segmentation. **Step 5: Select the optimal K** Since: the best model is:*

$K = 3$

Step 6: Interpret the three clusters *The three clusters can represent:*

Cluster 1 – High Performers

- High productivity
- High attendance
- High skill score

Cluster 2 – Average Performers

- Moderate productivity
- Moderate attendance
- Moderate skill score

Cluster 3 – Employees Needing Improvement

- Lower productivity
- Lower attendance
- Lower skill score

Step 7: Management benefits

The three-cluster model helps the company to:

1. Identify high-performing employees.
2. Recognize and reward good performance.
3. Identify employees requiring training.
4. Develop personalized improvement plans.
5. Improve workforce planning.
6. Monitor employee performance over time.

Step 8: Final justification

*$K = 3$ should be selected because it has the **highest silhouette score of 0.68**.*

*This indicates better separation and cohesion than $K = 2$ and $K = 4$. **Final Conclusion***

***$K = 3$ is the most suitable clustering model.** It provides meaningful employee groups and helps management make better decisions regarding rewards.*

